

The Musical Brain: An Introduction

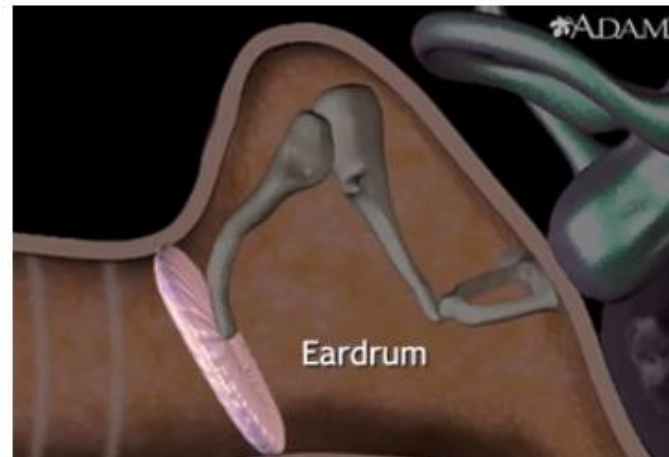
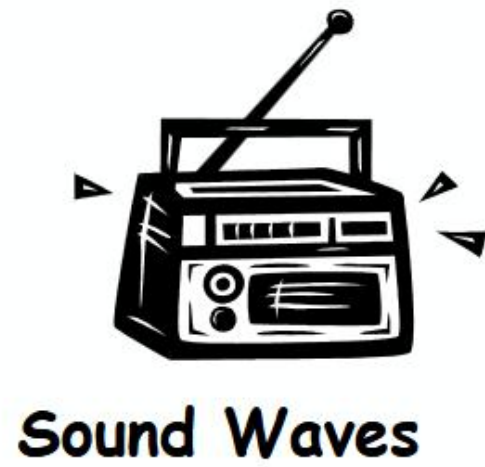
April 21, 2026

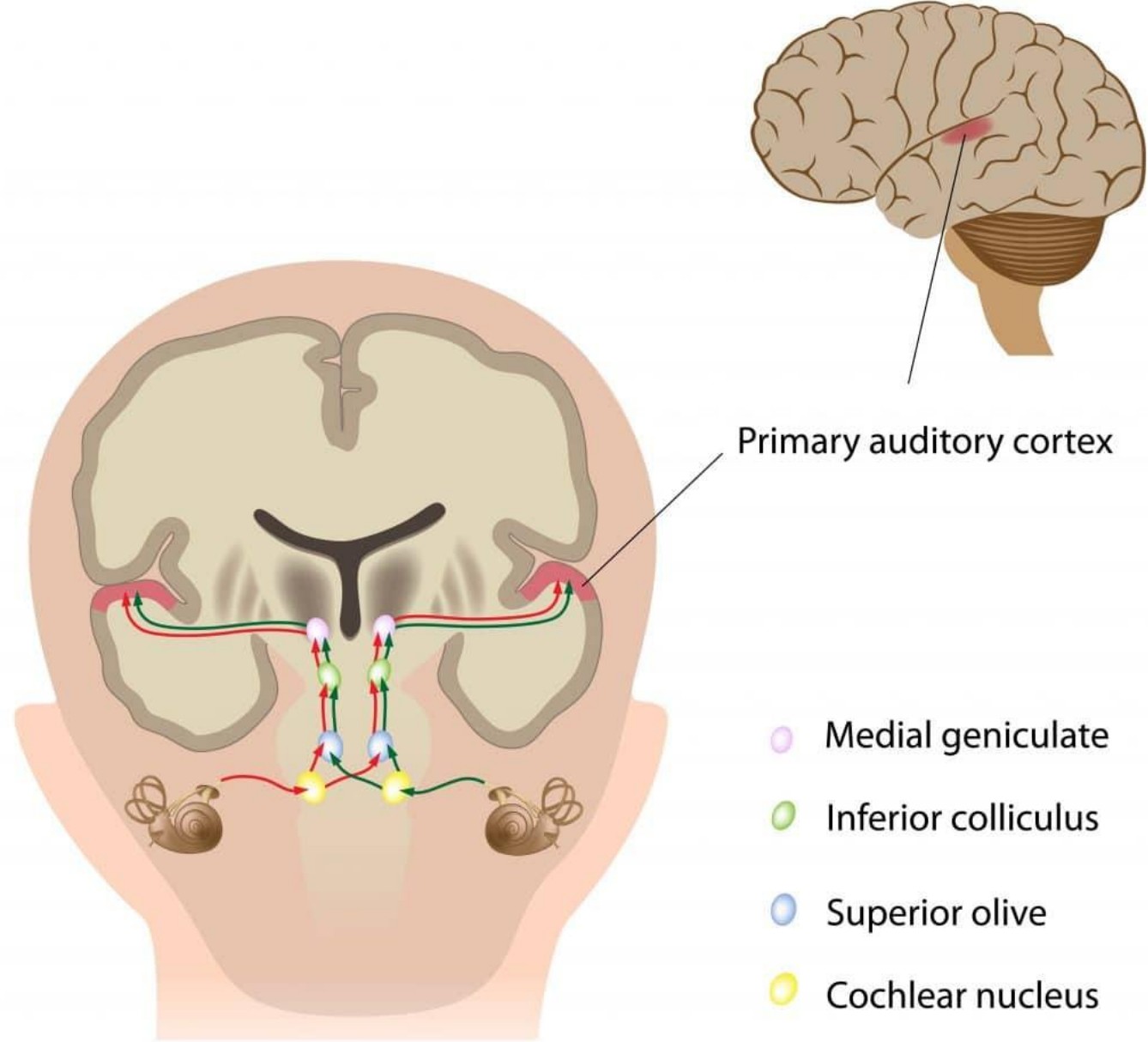


DeBÍ TiRAR Más FOToS – Bad Bunny

Auditory Scene







Music Listening



Used in
everyday life



Cognitive processes

- Sensory
- Attention
- Memory, and
- Emotional dimensions



Affordances

- Emotional regulation
- Sensory Modulation
- Reflection and Meaning-Making
- Pleasure
- Social Connection and Empathy

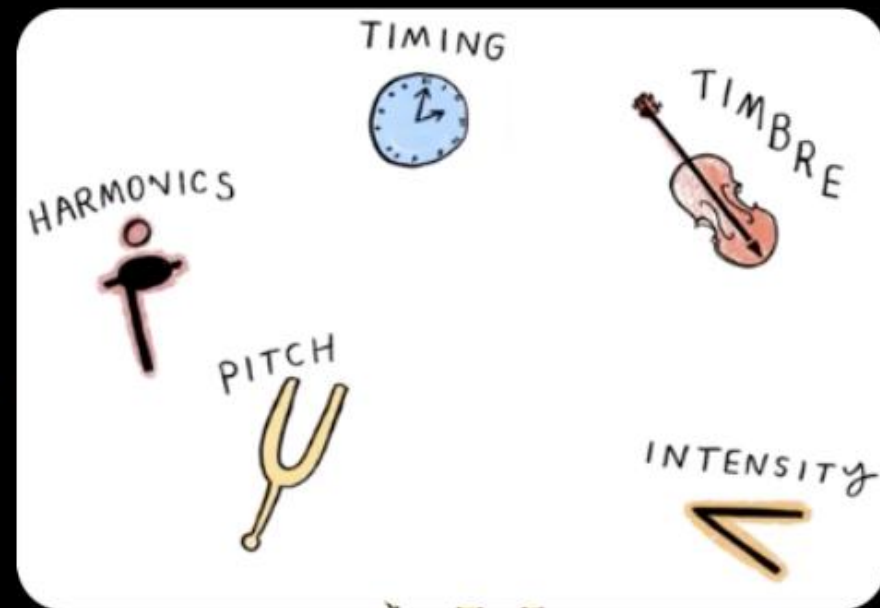
Naturalistic and
Real-world setting

“
Music is a mirror of the self. Music is an expression of who we are
emotionally, socially, and cognitively.”

Greenberg et al., 2015



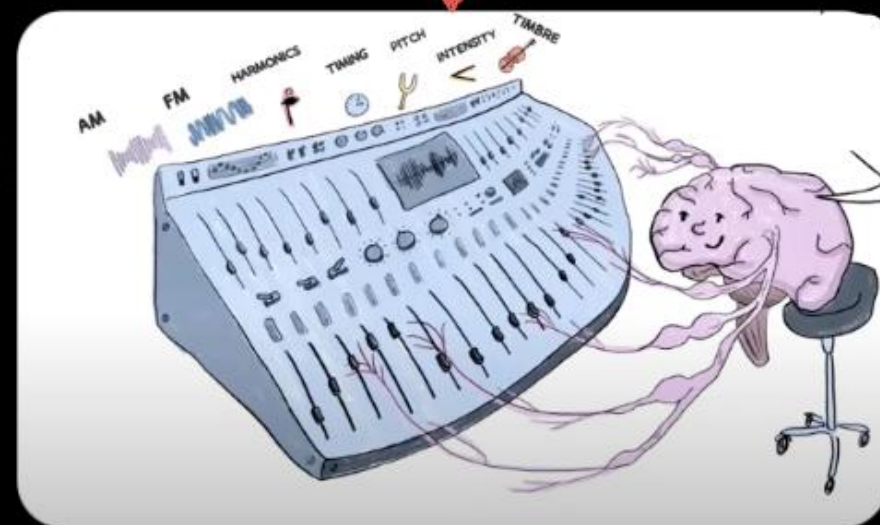
separate

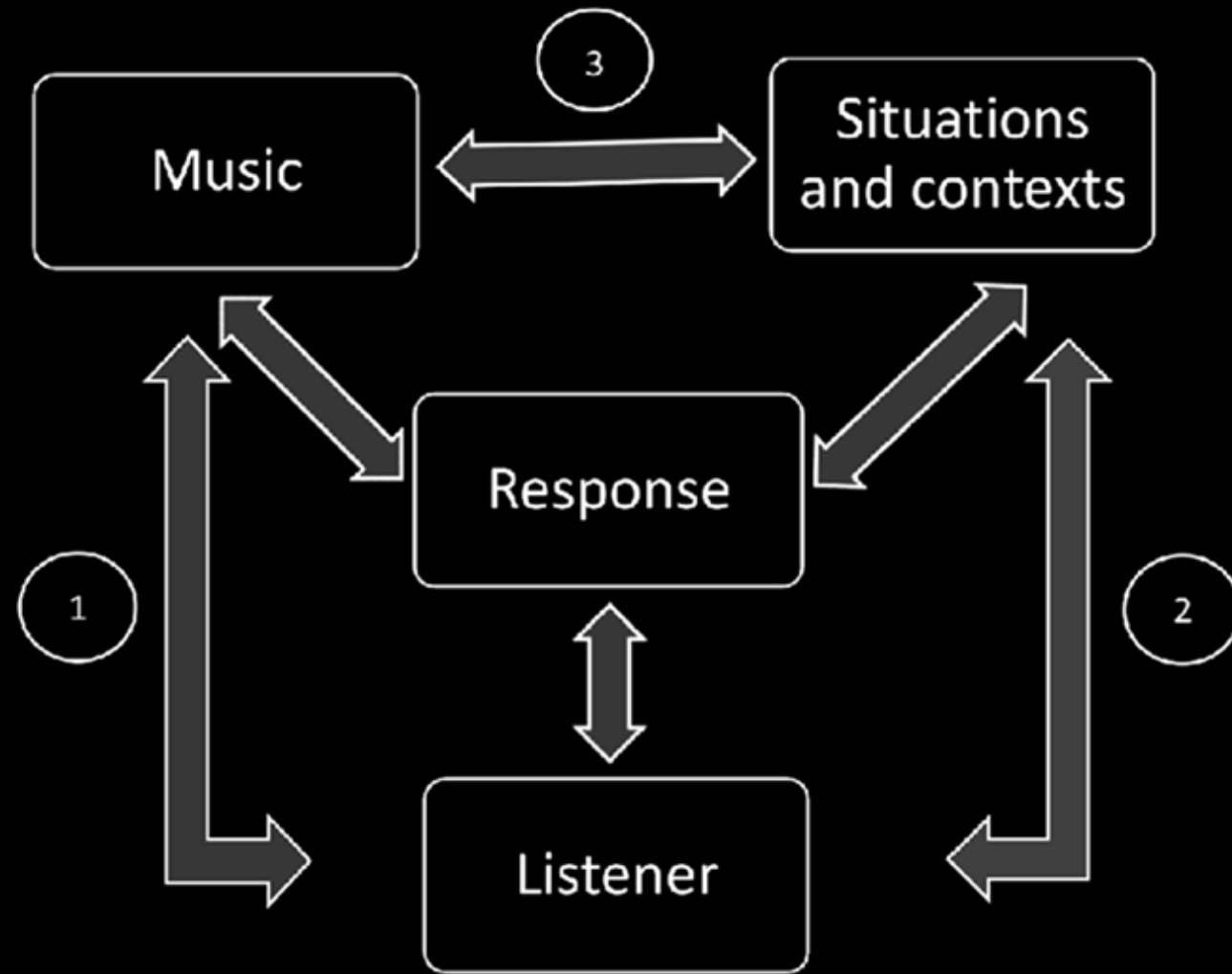


© Nina Kraus
Source: <https://www.youtube.com/watch?v=S7LUDeDGIcM>

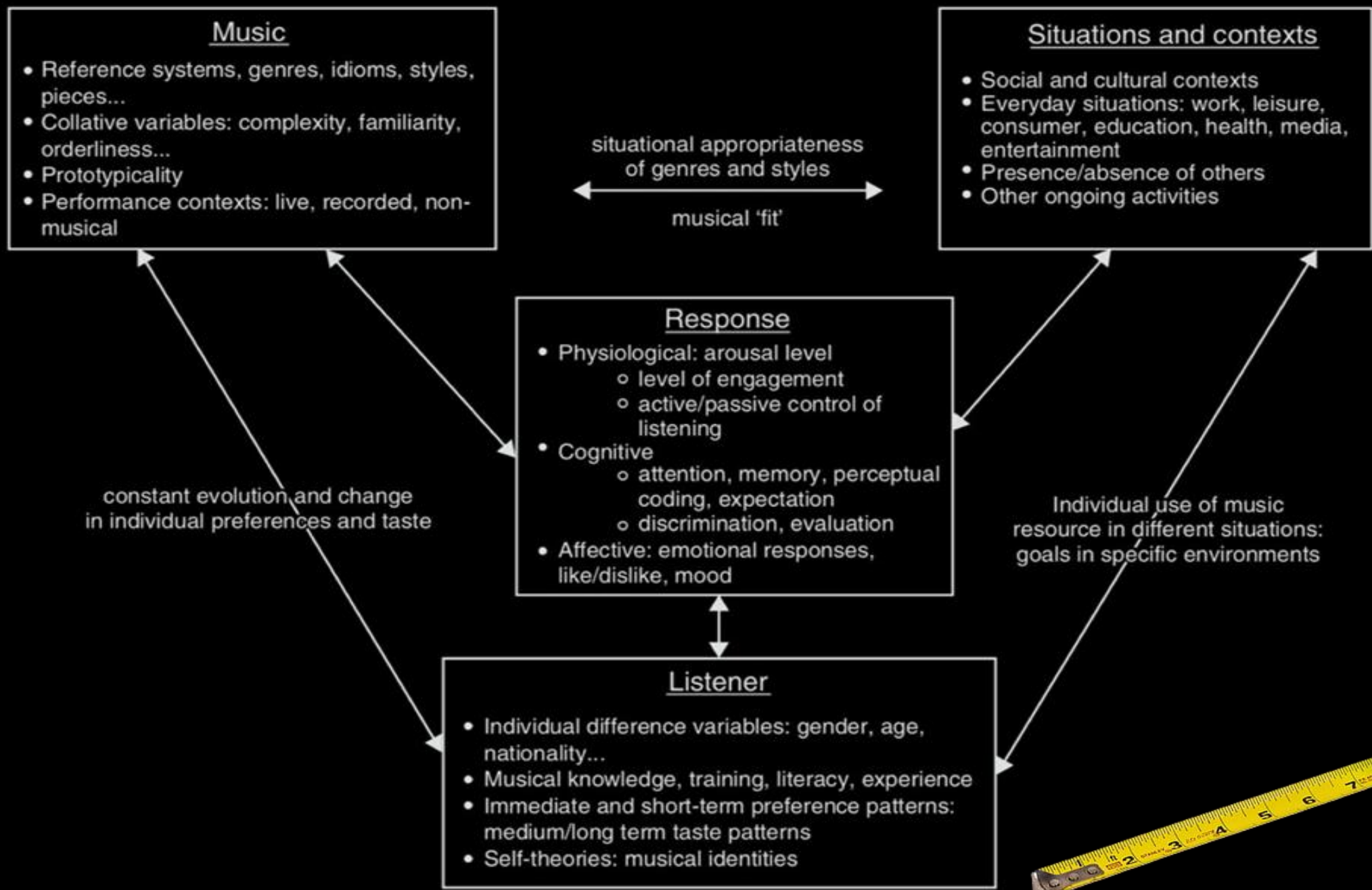
Melody
Harmony
Rhythm
Key
Identify
Memories
Emotions

integrate

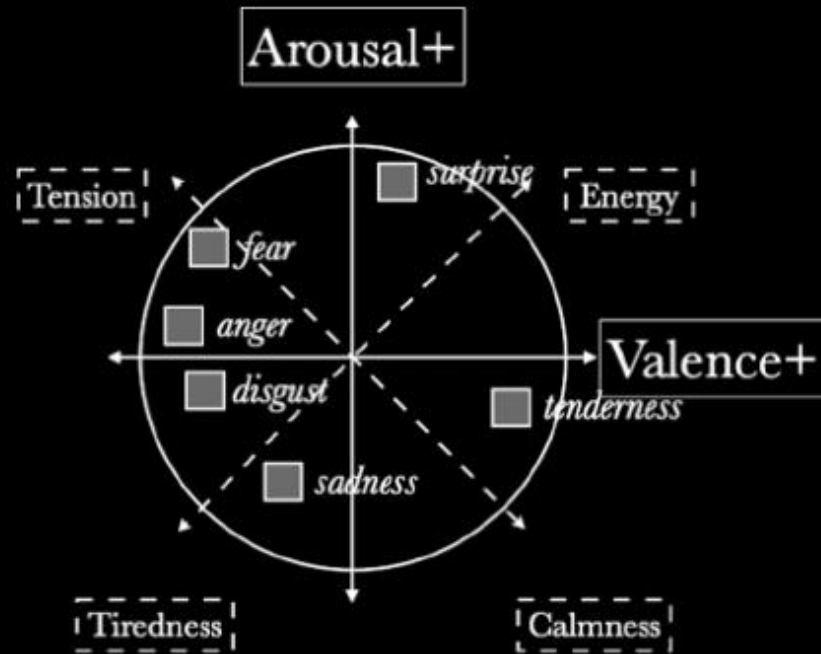




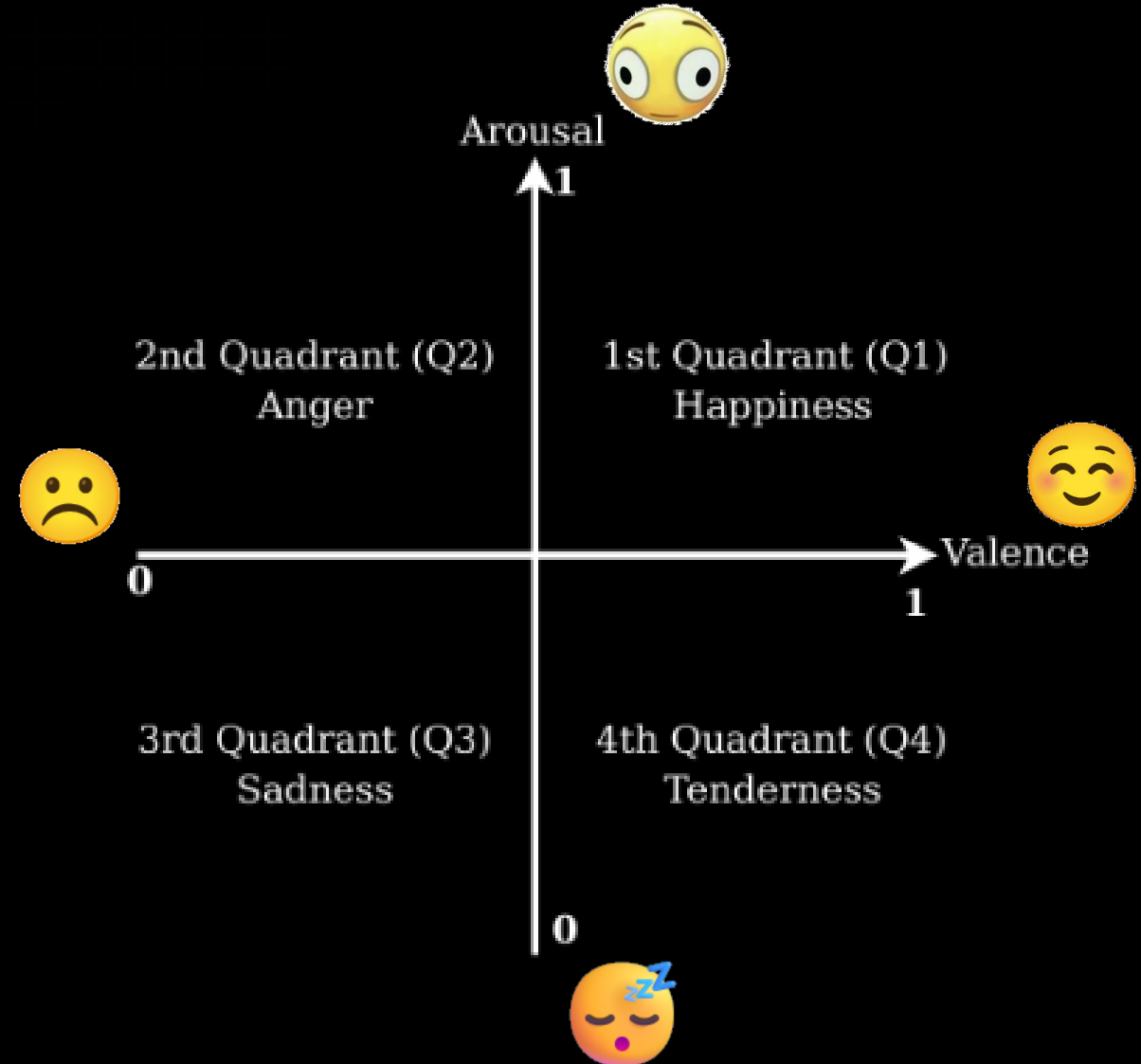
Reciprocal feedback model of musical response



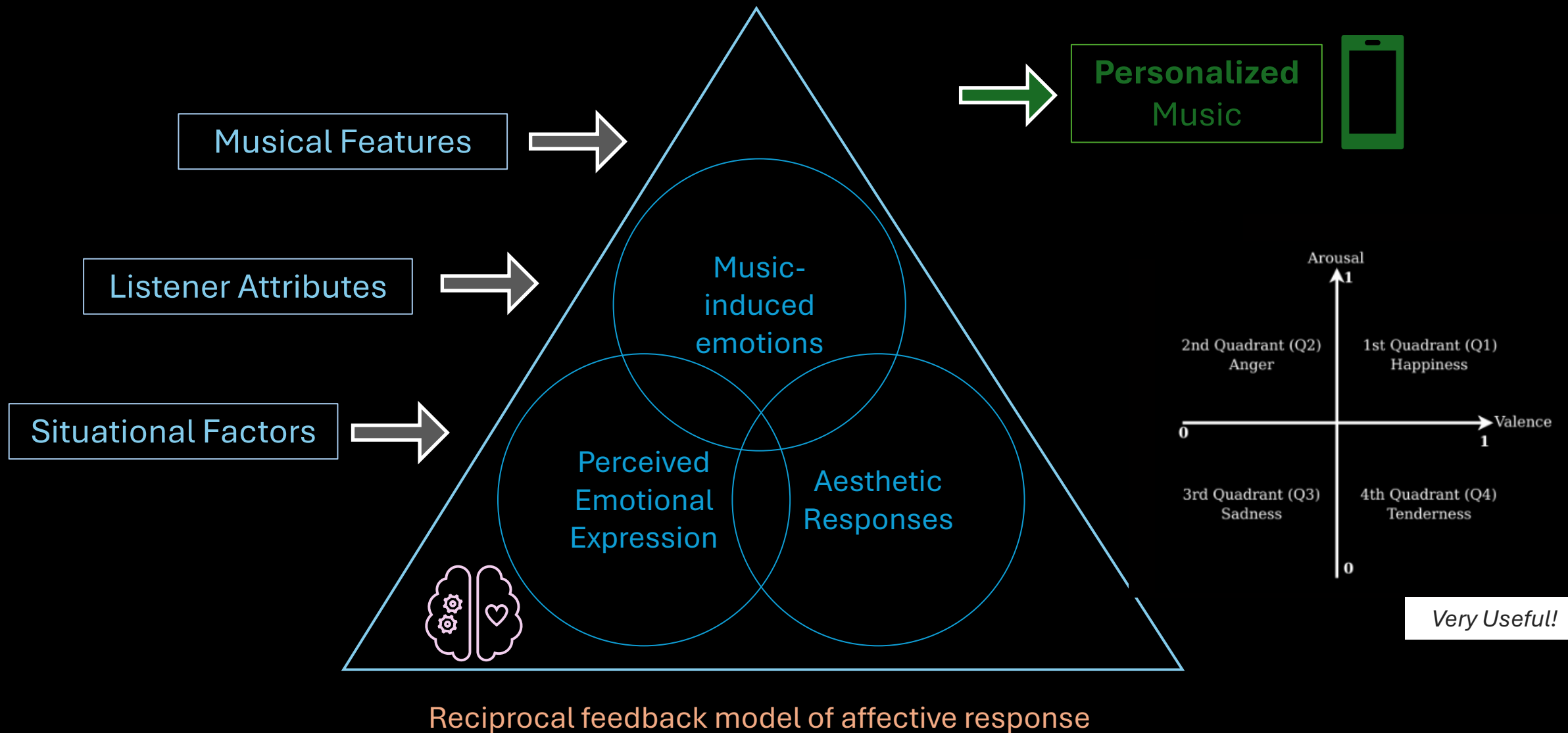
Quantifying (Emotion) Response



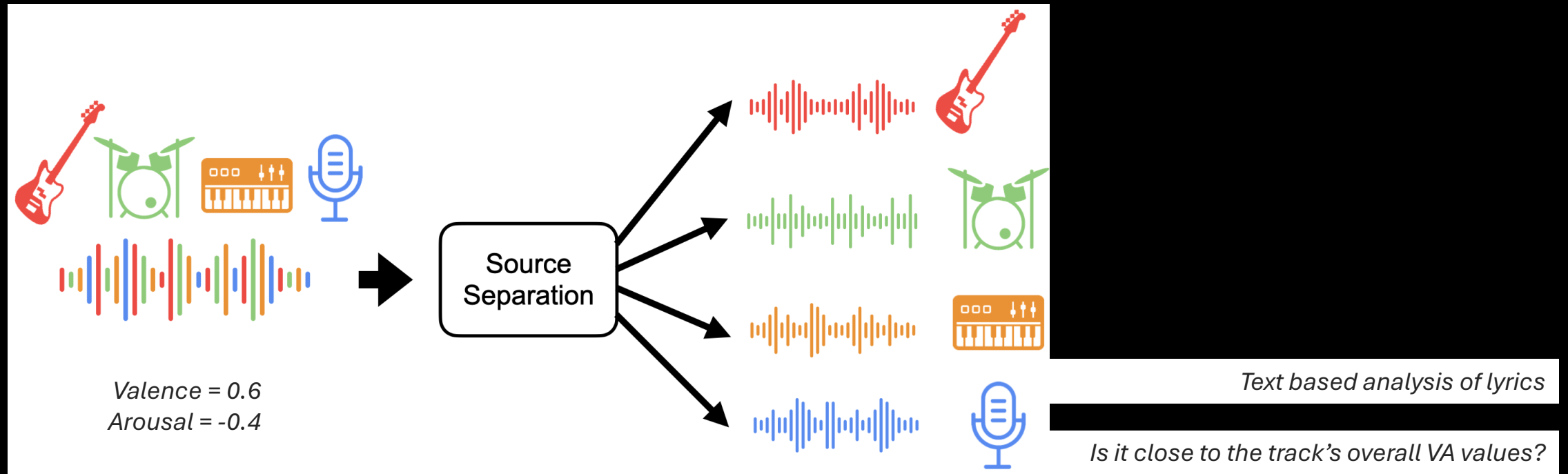
Russell's circumplex model ———
Thayer's model - - -
Basic emotion terms ■



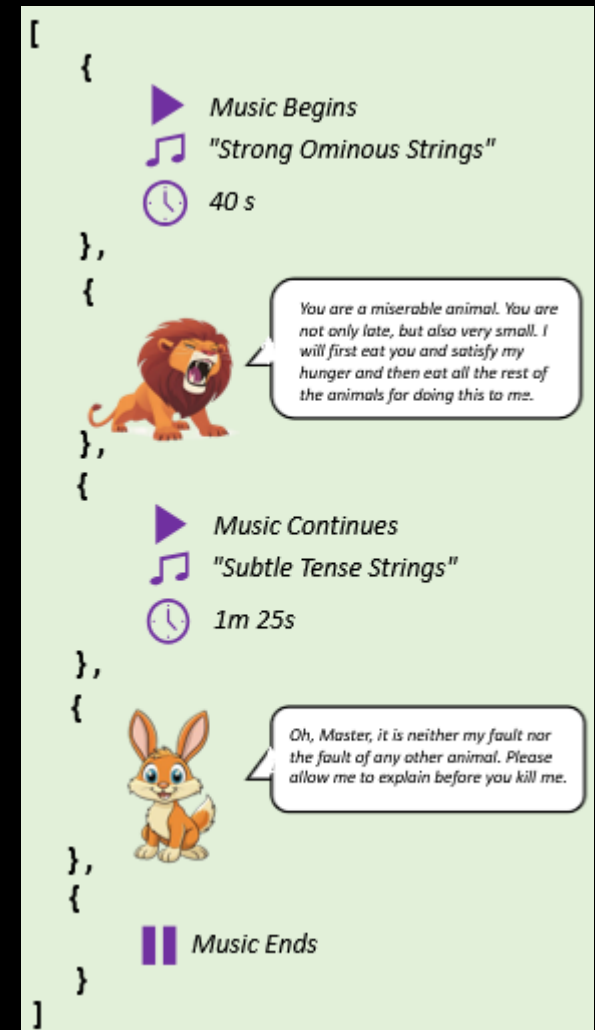
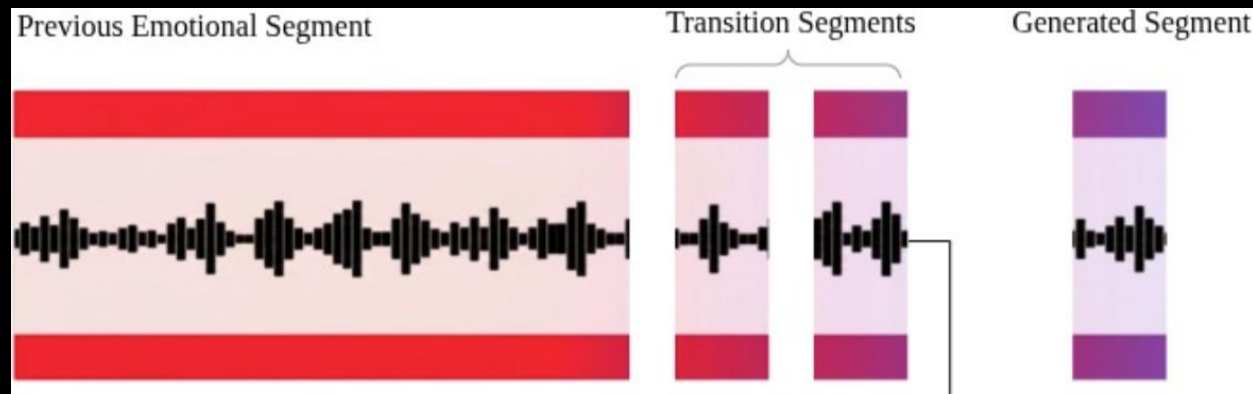
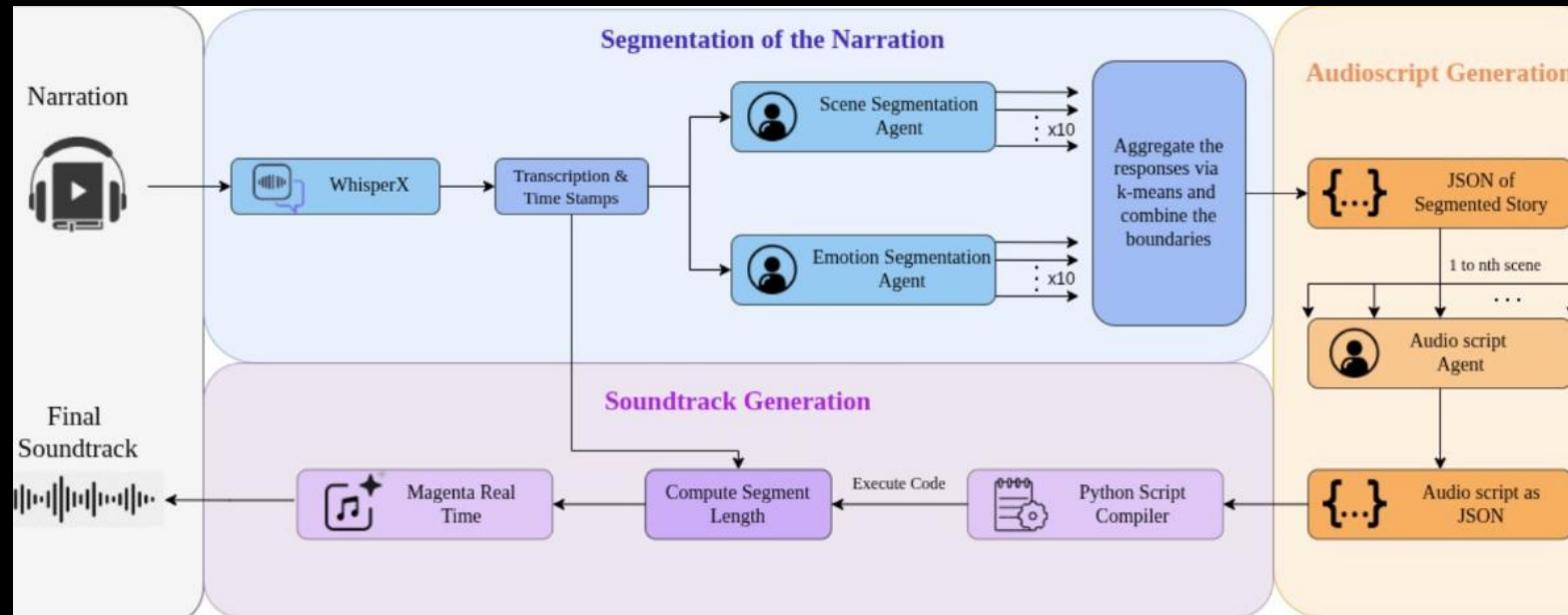
Differences in our emotional responses



Does the sadness come from the lyrics or the music?



How do I make reality more exciting?

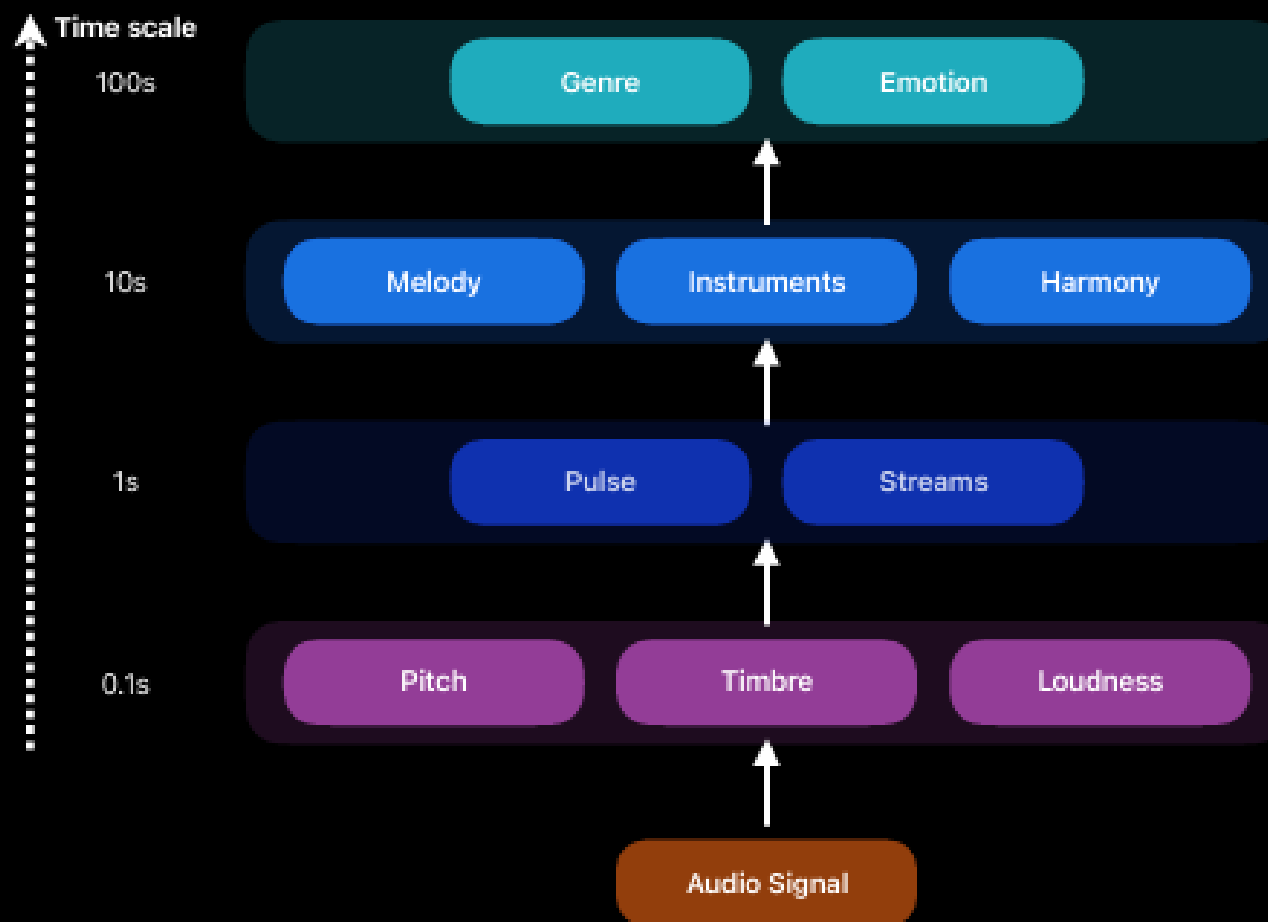


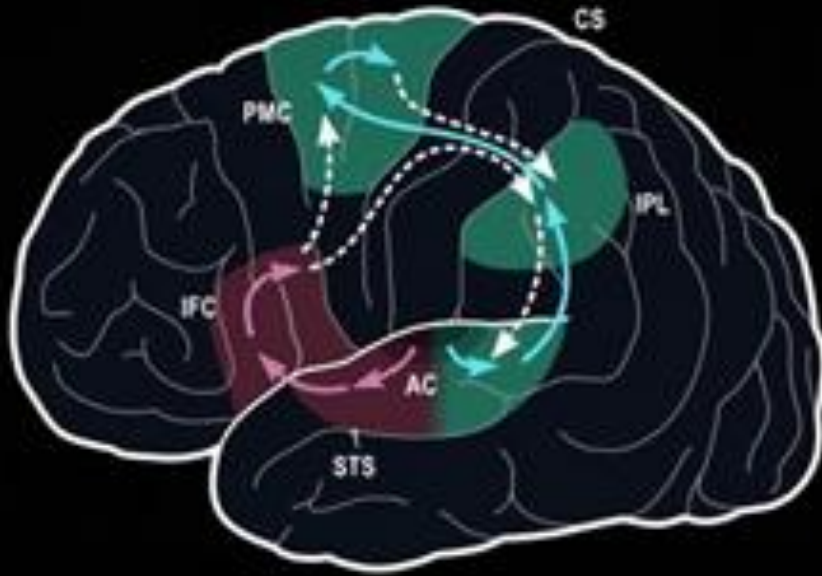
What makes music recognizable?

Jazz



Hip Hop



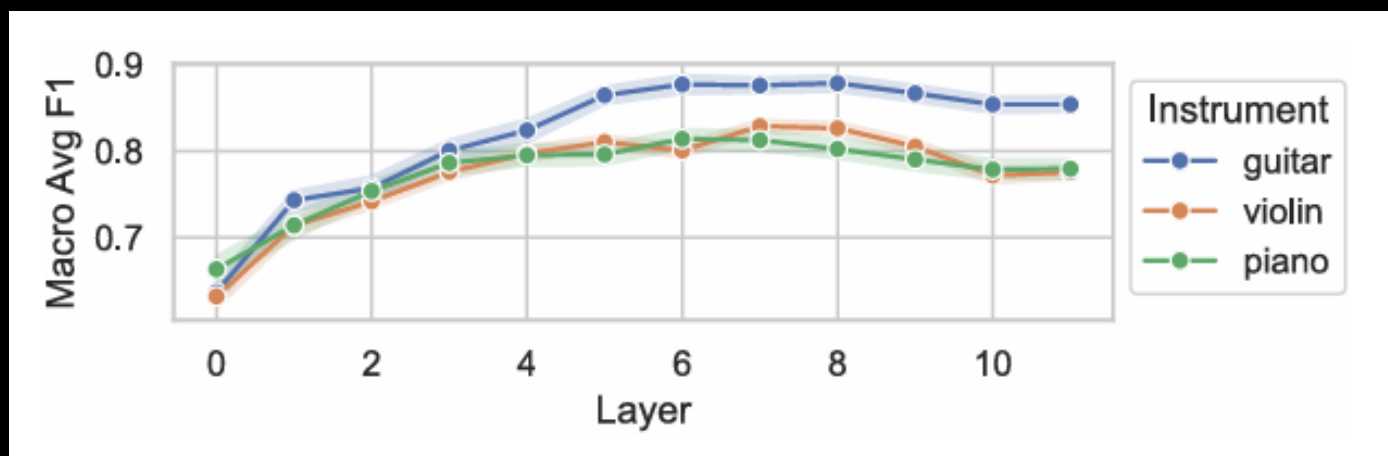
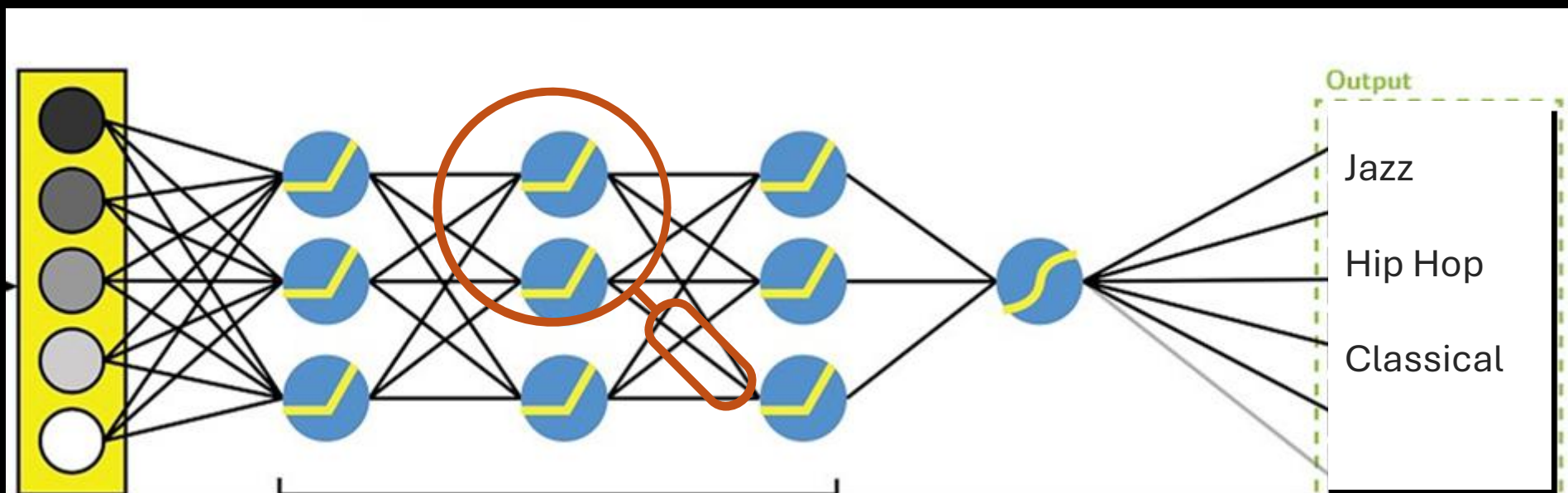


(b) Dorsal and ventral pathways for auditory processing in the human brain. Figure from Rauschecker [63].^a

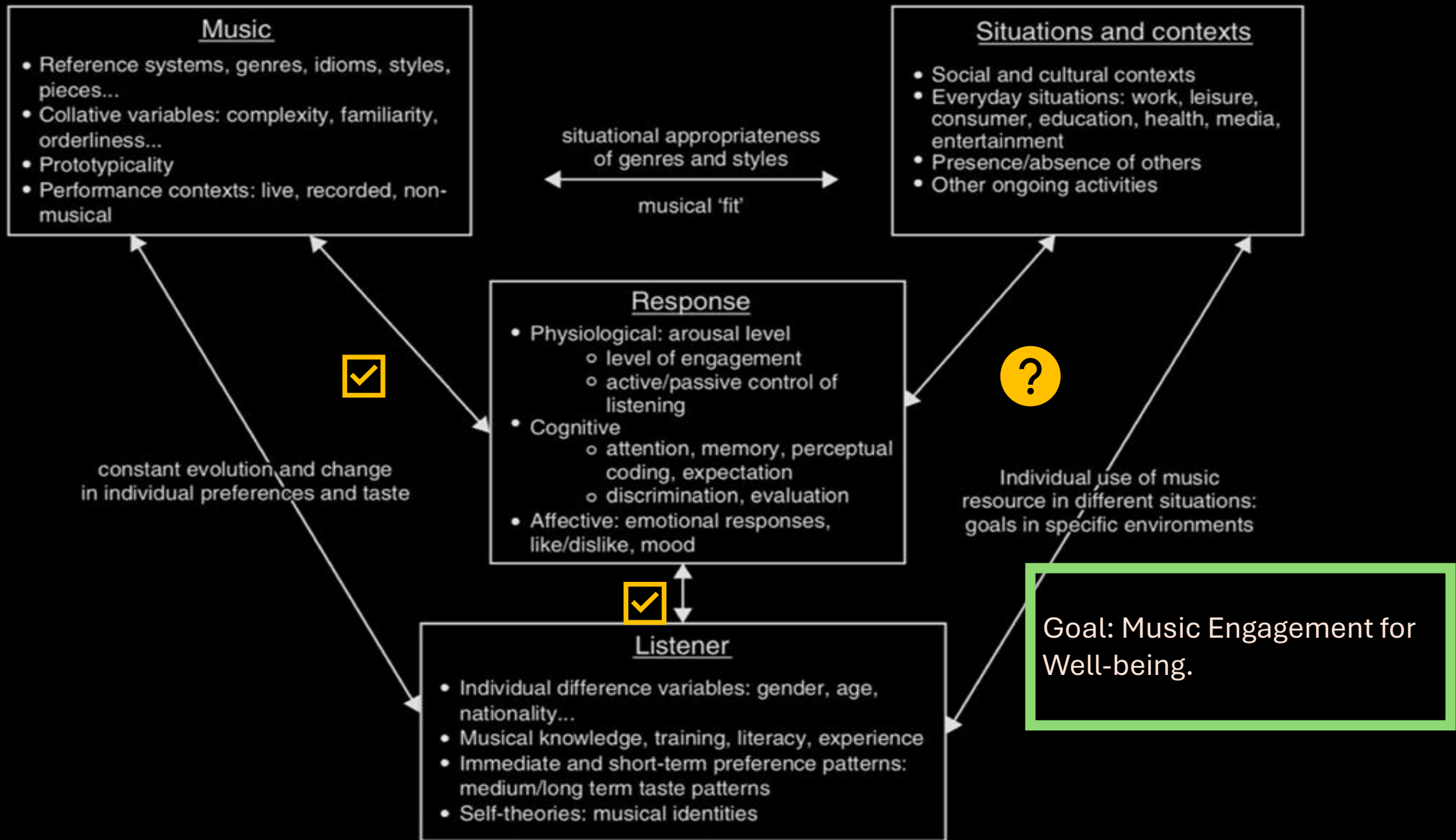
^aAC: auditory cortex; STS: superior temporal sulcus; IFC: inferior frontal cortex; PMC: premotor cortex; IPL: inferior parietal lobule; CS: central sulcus.



Train a neural network on genre classification and probe its layers



Credit: Pratyaksh Gautam



Healthy Music Engagement in Listeners

- music-based interventions for recovery and alleviating symptoms in neurological and clinical disorders ^{a, b}
- create positive emotional states and increase energy levels ^{a, b}
- promote feelings of well-being ^{b, c}
- helps in regulating emotions ^{a, b, d}



^a Reybrouck, 2020; ^b Gustavson et al., 2021; ^c Alluri et al., 2022; ^d Tan et al., 2024

Unhealthy/Maladaptive Music Engagement

- associated with Internalizing Disorders ^{a, b}
- negative emotionality and increased odds of experiencing depressive symptoms ^{a, b, d}
- susceptibility to negative emotional contagion ^c
- difficulty managing emotions when in distress ^d

Context Matters!

“Health is socially produced and environmentally-mediated”

- Denora 2007

(Health and music in everyday life – a theory of practice)

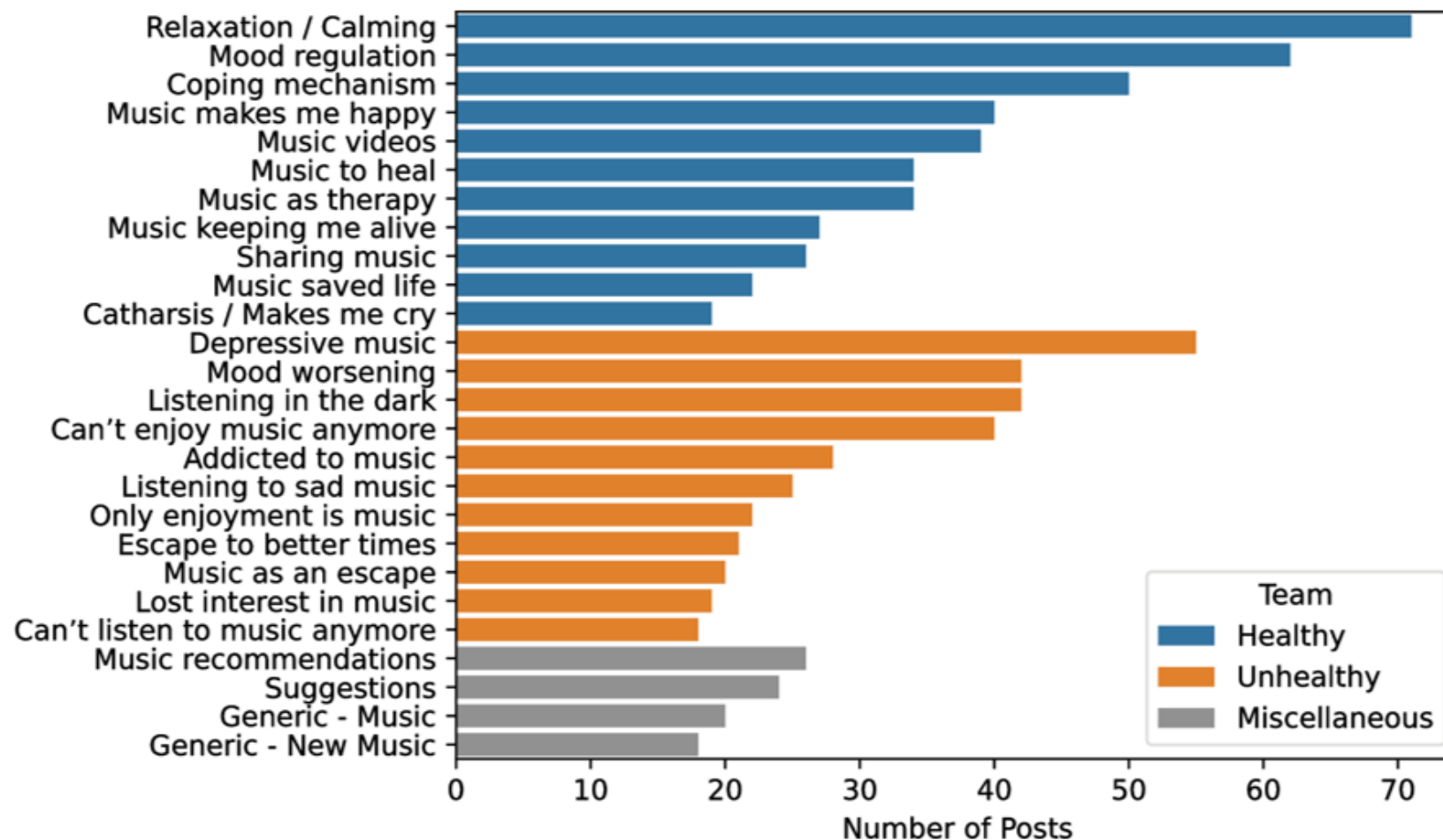
^a Gustavson et al., 2021; ^b Kanagala et al., 2021; ^c Alluri et al., 2022; ^d Tan et al., 2024

Digital Footprints





Social Media: Depression + Music





adjectives, adverbs embeddings

Word Emotion Induction Model

Word	Valence	Arousal	Dominance
sunshine	8.1	5.3	5.4
terrorism	1.6	7.4	2.7
orgasm	8.0	7.2	5.8

Table 2: Three sample entries from Warriner et al. (2013). They use 9-point scales ranging from 1 (most negative/calm/submissive) to 9 (most positive/excited/dominant).

- semi-structured data (e.g. JSON)
- meta-data
 - lyrics
 - acoustic features
 - tags

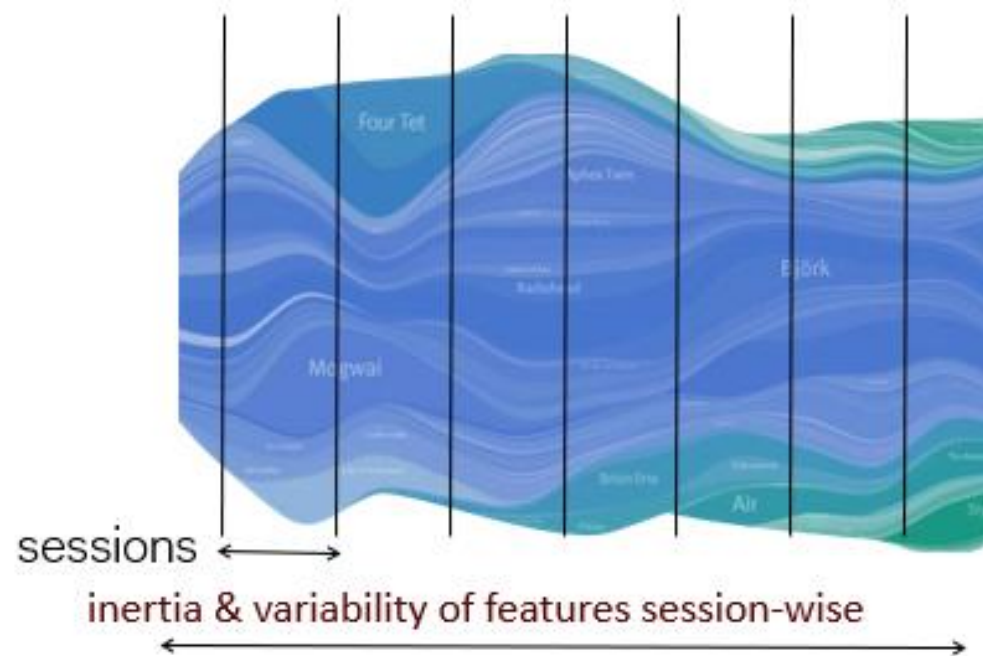


Eerola, T (2022), MMMT Workshop @ISMIR 2022

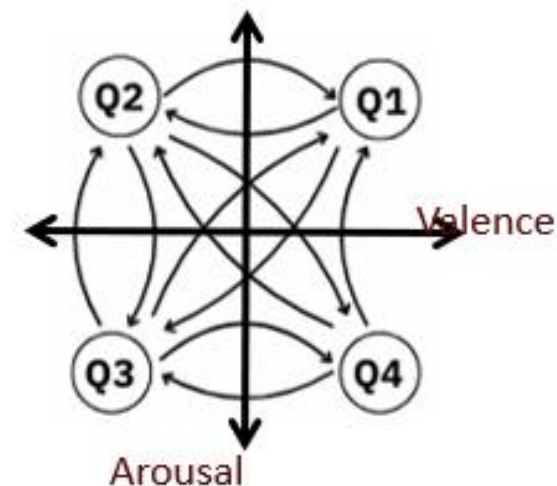




- semi-structured data (e.g. JSON)
 - meta-data
 - lyrics
 - acoustic features
 - tags
- feature engineering
- static vs dynamic



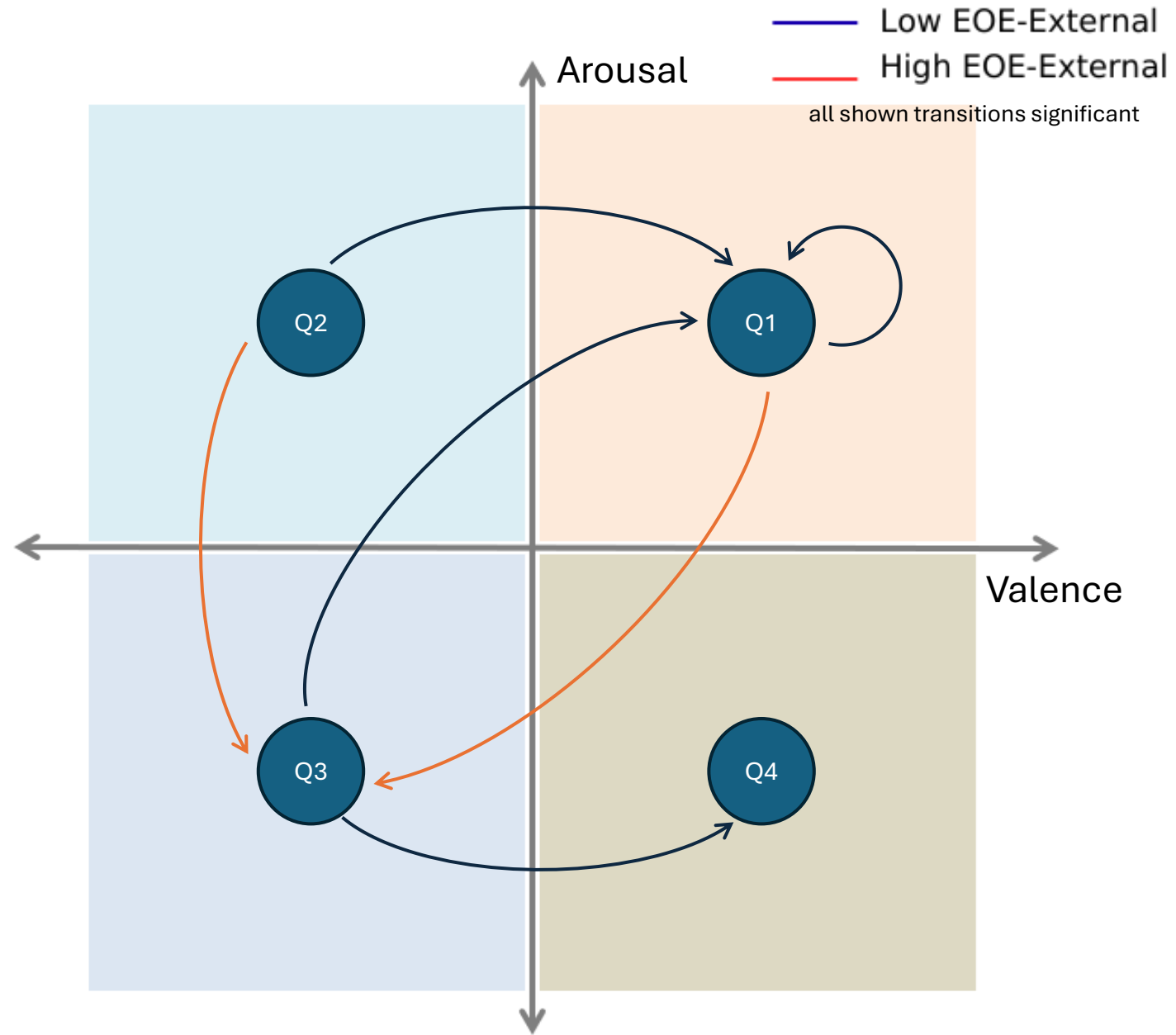
emotion transition probabilities



+ Socio Affective Markers

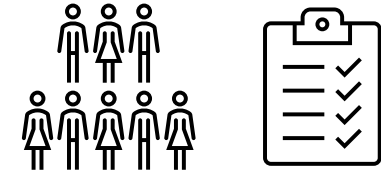
What are the markers associated maladaptive music use across different trait-level groups in the listening history?

- Do highly sensitive persons have different music listening patterns than individuals at risk of depression?
- What are musical markers of rumination and avoidance for individuals showing maladaptive music engagement?



EOE External Transition Matrix

Is music use driving well-being outcomes?

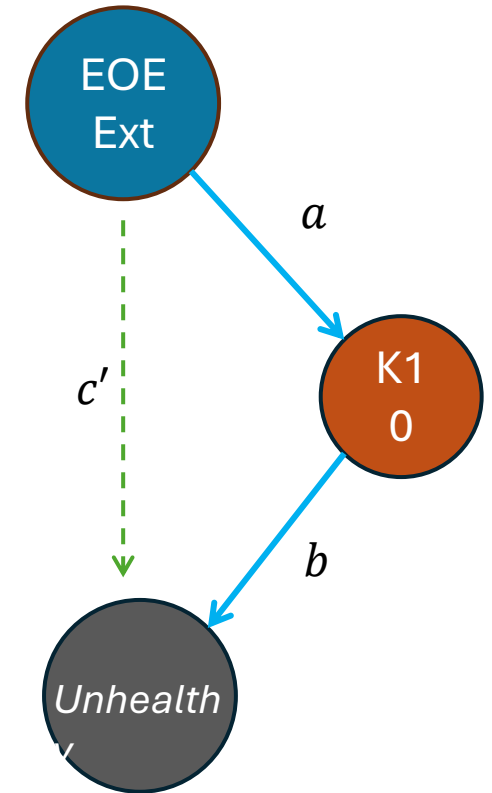
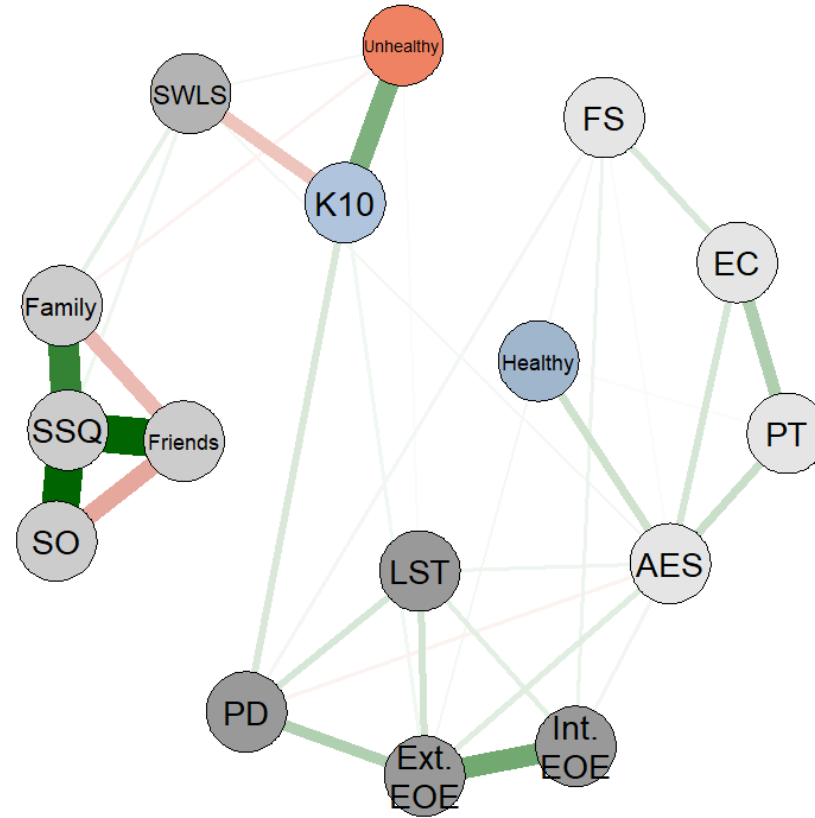


- Take large scale cross-sectional data
- participants complete an online survey consisting of six standard questionnaires:
 - › HSPS^d and HUMS^f
 - › **Kessler Psychological Distress Scale (K10)**^e – anxiety and depression symptoms
 - › **Interpersonal Reactivity Index (IRI)**^b – empathy
 - Perspective Taking (PT), Fantasy Seeking (FS), Empathic Concern (EC), and Personal Distress (PD)
 - › Satisfaction with Life Scale (SWLS)^c – overall life satisfaction
 - › Social Support Questionnaire (SSQ)^a – perceived social support from family, friends, and significant others

^aSarason et al., (1983); ^bDavis, (1983); ^cDiener et al., (1985); ^dAron & Aron, (1997); ^eAndrews & Slade, 2001; ^fSaarikallio et al., (2015);

Network Analyses

- Make sure of reliability and consistency of psychological variables (e.g., traits, symptoms, mediators)
- Have them as **interacting nodes**, capturing their *conditional dependencies*.
- Fit directed acyclic graphs to see if data fits a model of one's hypothesis.



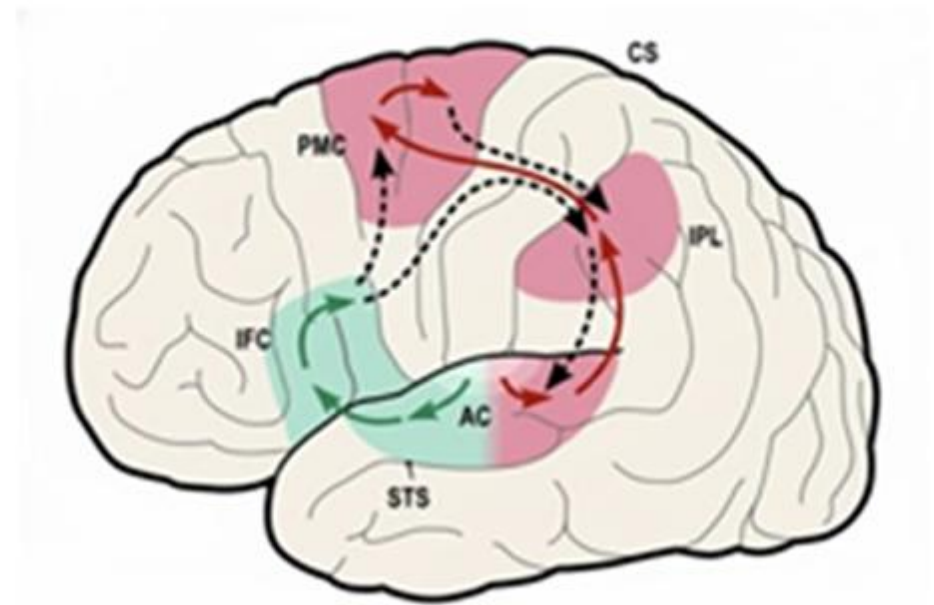
Experience Sampling



- Add physiological and behavioural measures during music listening:
 - Heart Rate Variability;
 - Goal relevance;
 - Hedonic Shift
- Define Groups across which we expect to see differences.
- Conduct Longitudinal Study

Then?

We can go back to
where it all happens...



(b) Dorsal and ventral pathways for auditory processing in the human brain. Figure from Rauschecker [63]. ^a

^aAC: auditory cortex; STS: superior temporal sulcus; IFC: inferior frontal cortex; PMC: premotor cortex; IPL: inferior parietal lobule; CS: central sulcus.



EVOLUTION OF MUSIC



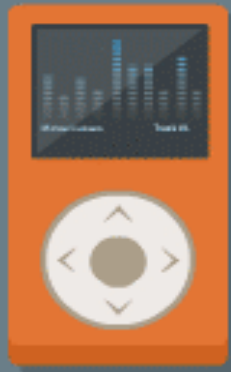
VINYL RECORDS
PETER CARL GOLDMARK
1887



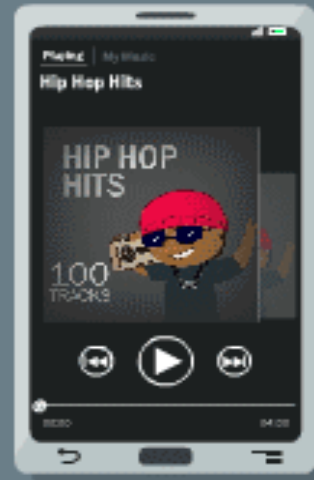
CASSETTE TAPES
PHILIPS COMPANY
1962



AUDIO CD
JAMES RUSSELL
1982



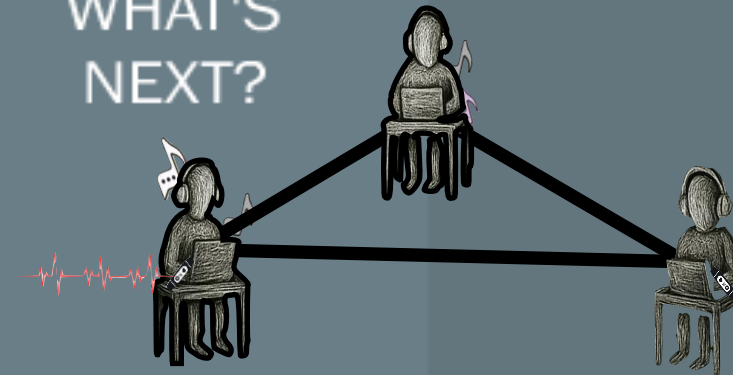
MP3 Player
Kane Kramer
1998



Cell Phones
Samsung Uproar
2001



WHAT'S
NEXT?



Assignment

- Pick two tracks:
 - one of your current favorite tracks and
 - one of favorite tracks from three to four years ago.
- Listen for three distinct streams of audio within the two songs (it can be an instrument, melody, or rhythm).
- Document which stream grabs your attention the most and which one the least (you can pick a segment of the songs for this).
- Document your assessment of felt VA values for entire song. Does it change?
- Identify one Listener Attribute unique to you that relates to your answers.

